

# Deep Learning Fundamentals

Deep learning is a subset of machine learning that uses neural networks with multiple layers (deep neural networks) to model complex patterns.

Key Concepts:

1. Neural Networks - Inspired by biological neurons
2. Backpropagation - Algorithm for training networks
3. Activation Functions - ReLU, Sigmoid, Tanh
4. CNNs - For image processing
5. RNNs - For sequential data

Popular frameworks: TensorFlow, PyTorch, Keras